

9.1 What is a data dashboard? ↕

□ Activity: Dashboard analysis

Choose a dashboard from the gallery of dashboards available at the following webpage. As you look at this dashboard, take time to consider and make notes on:

- whether this dashboard is appealing to you
- whether the use of pre-attentive attributes is helpful for the audience
- whether the use of Gestalt principles is good
- whether data-ink ratio has been used well
- whether eye movement is minimised.

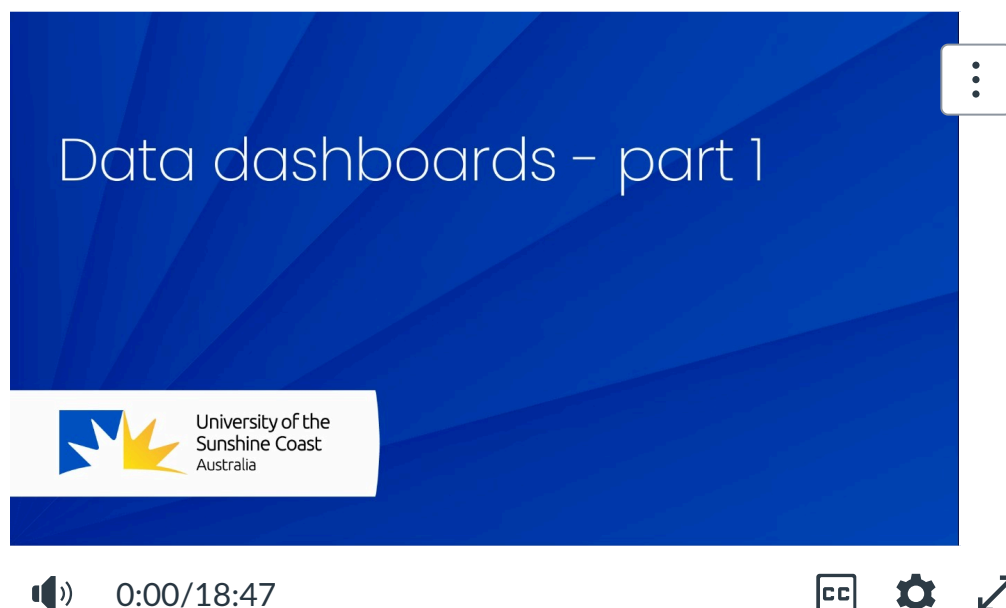
Explain your reasoning and bring your notes to the workshop.

Infocaptor. (n.d.). [Realtime dashboards](https://www.infocaptor.com/dashboard-gallery) ↗ (<https://www.infocaptor.com/dashboard-gallery>).

□ Watch: Data dashboards (part 1) [18.47 mins]

Watch the video below, which explains the principles of effective data dashboards and describes various dashboard types and taxonomies.

This is useful as it provides a fundamental understanding of the range of ways in which effective data dashboards can be designed and used to facilitate better decision making.



Transcript

<https://learn.usc.edu.au/courses/35242/files/3138523?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Data dashboards \(part 1\)](https://learn.usc.edu.au/courses/35242/files/3168287?wrap=1)
<https://learn.usc.edu.au/courses/35242/files/3168287?wrap=1>

Just like a vehicle dashboard – which provides the driver with information about variables such as oil pressure, fuel level and speed etc. – a data dashboard provides people working at various levels within a business the information they need to make be able to make decisions. Wherever business decisions are being made, appropriate information is needed to support the decision-making process. This is sometimes known as the *moment of value*.

All business managers have similar information needs in order for them to effectively and efficiently manage and operate their businesses. For example, they may need to know their organisation's financial position, levels of inventory, pending orders and metrics around customer service. The values of these variables is known as *key performance indicators* (KPIs). The KPIs will differ depending on the type of organisation.

Data dashboards are visualisations tools that provide decision makers with timely information on the KPIs relating to their jobs/organisations. Many decision makers rely on the multiple outputs that data dashboards provide, outputs which may be updated in real time. Organisational KPIs are aligned with an organisation's goals. They allow the continual monitoring of current and potential future performance. Data dashboards, and the visualisations therein, are usually designed for (a) particular purpose/s. The data from which the visualisations are developed can be drawn from multiple sources.

A data dashboard should convey meaning to its user and relate to the decisions that that user needs to make. It should present the KPI information about an aspect of a business's operations on a single screen. This allows a user to review the current situation quickly and accurately "at a glance". The need to scroll either vertically or horizontally detracts from the ability to gain a full understanding of the situation. Therefore, in situations where scrolling seems to be unavoidable, it is better to create multiple dashboards of related KPIs.

□ Watch: Five tips for building better data dashboards [5.59 mins]

This video is useful as it reiterates the importance of knowing your audience, the story that you want to tell, and your data and the data sources. It also advises having a story plan and knowing how to use the tools needed to create the most appropriate visualisation.

As you watch, consider:

- How does knowing your audience influence the composition of your data dashboard?
- Why is this important?



Source: © Measure School (2017).

There are many different types of application where data dashboards are useful. These include (but are not limited to):

- crime dashboards
- donor dashboards
- health/personal fitness dashboards
- human resource dashboards
- investment dashboards
- manufacturing dashboards
- marketing dashboards
- school performance dashboards
- technical support dashboards.

□ Activity: The cost of cybercrime

Read the following report from 2019. As you read, take time to consider and make notes on:

- how you would approach creating a data dashboard to represent this data
- what techniques you would use, and why.

Ensure that you can justify your thinking and bring your answer to the workshop.

Ponemon Institute & Accenture. (2019). [The cost of cybercrime](https://www.accenture.com/content/dam/accenture/final/a-com-migration/pdf/pdf-96/accenture-2019-cost-of-cybercrime-study-final.pdf) 
(<https://www.accenture.com/content/dam/accenture/final/a-com-migration/pdf/pdf-96/accenture-2019-cost-of-cybercrime-study-final.pdf>). Accenture.